Week 1, part 1 of 3

Machine Learning Engineering

COMP3110

Week 1

- **1.1 Course intro**:
- **1.2 Law of the instrument**
- **1.3 Model selection, routing, and gateways**

1.1 COURSE INTRO

What is Machine Learning Engineering?

MLE vs ML

- **ML** is the algorithms, maths and models. See courses like COMP9417.
- **MLE** is an application of ML, responsible for building a working system. See the job ads below in notes.
- **AI Engineer** also encompassed in 3110, basically LLMs in MLE.
- **COMP3110** is about building those ML/AI systems, end to end.

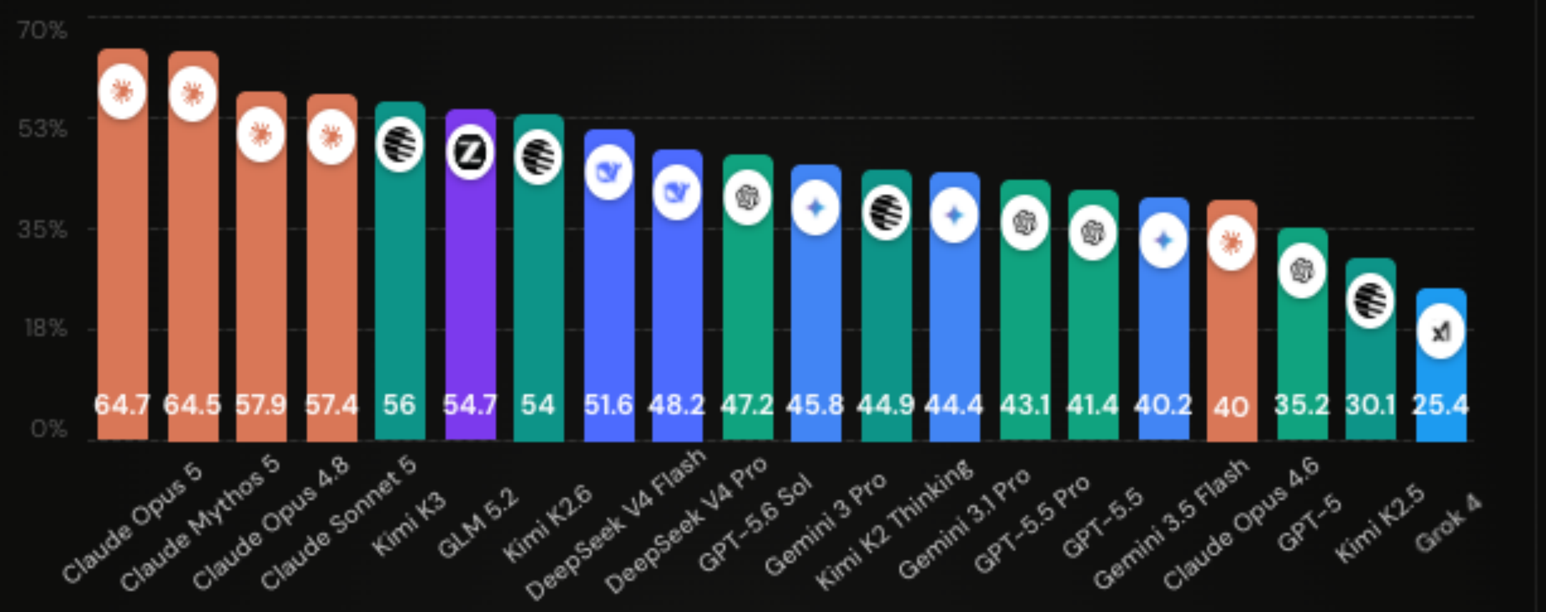
The focus of COMP3110

Choose a problem, build a system, measure its impact, and deploy it.

LLMS VS ML VS AI

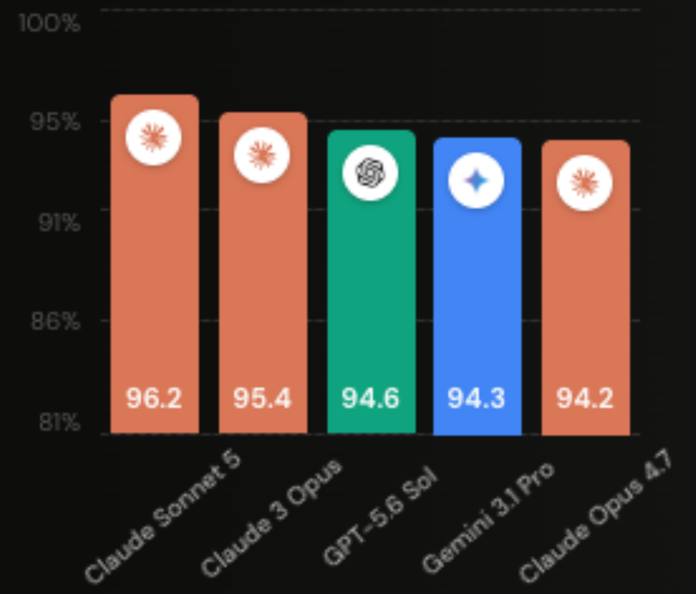
Be flexible with your models

Compare models on your task, with a dated evaluation and a cost budget.

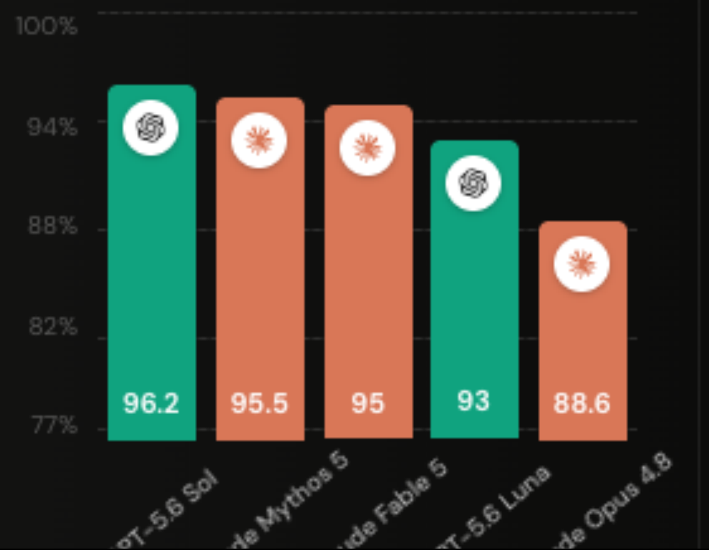


Top models per tasks

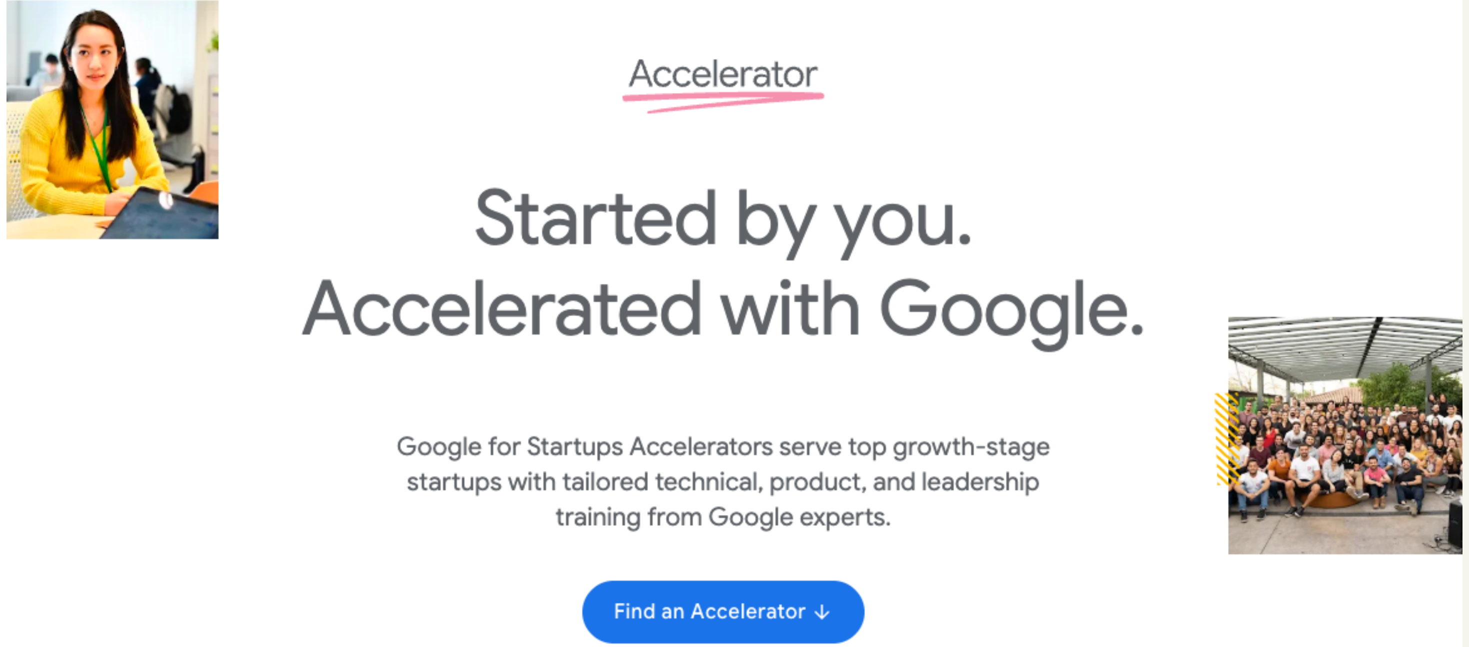
Best in Reasoning (GPQA Diamond)



Best in Agentic Coding (SWE Bench)



COMP3110 is an accelerator



The term at a glance

- **Onboarding** (Weeks 1-2): form teams, pitch a problem, get started
- **Sprint 1** (Weeks 3-5): a prototype that runs end to end and ideally produces a metric.
- **Sprint 2** (Weeks 7-9): metrics, evals, and improving metrics.
- **Sprint 3** (Weeks 10): polish, Demo Day prep.
- **Demo Day** (Week 11): a short, live presentation of your deployed system.

What to expect

- An emphasis on the problem you choose to solve.
- A non-traditional lecture and content structure.
- Lots of support for you and your teams, including industry mentors.
- A lot of self-learning, so you can meet your own requirements.
- An in-person Demo Day to present your project.

What not to expect

- Highly specified tasks.
- Someone handing you the dataset, projects, metrics, or the solutions.

The course site

Everything (logistics, timetable, the accelerator, projects, mentors) lives here.

Mentors

Mentors are available throughout term to help with your MLE project.

Some guidelines:

– Mentors volunteer their time. Treat it with respect.
– Bring a specific problem that matches the mentor's skills.
– Be respectful, polite, grateful, and mindful of their time.

Assessment

- **Portfolio of Tasks (60%)**: evidence across onboarding and accelerator tasks, including individual contributions.
  - Broken up into Accelerator tasks (30%) and Technical tasks (30%)
- **Final Project (40%)**: your final deliverable, including Demo Day performance and communication.

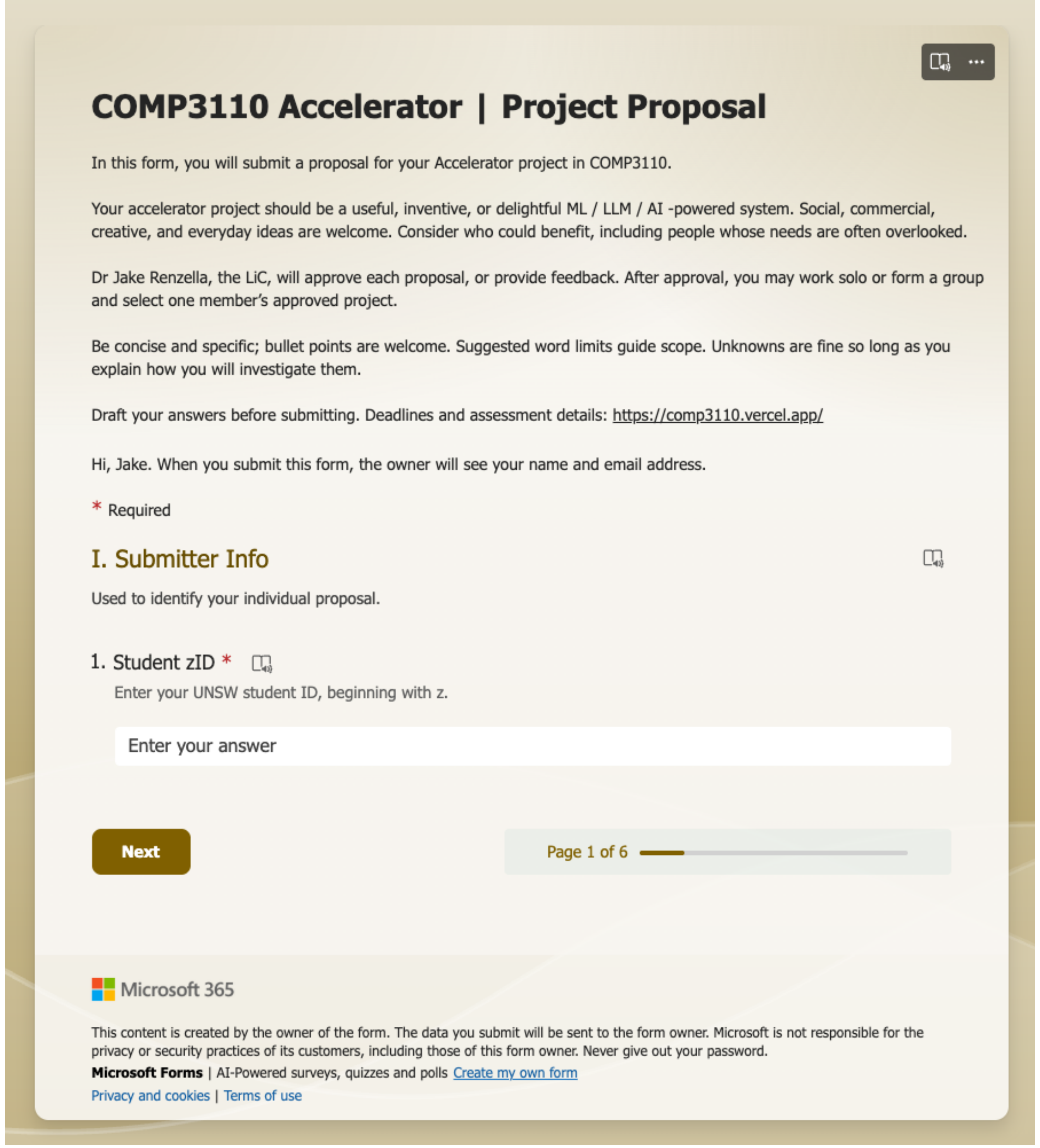
Guest lectures

- **Technical and non-technical**: A variety of topics from leaders in the field.
  - Let me know you have requests.
- **Scheduled on the Syllabus page**.
- **Attendance & Rules**: Most guest lectures will follow the Chatham House Rule. Important to respect this.[1]

1. "When a meeting, or part thereof, is held under the Chatham House Rule, participants are free to use the information received, but neither the identity nor the affiliation of the speaker(s), nor that of any other participant, may be revealed." Chatham House. https://www.chathamhouse.org/about-us/chatham-house-rule ↩

PITCHING A GOOD PROJECT PROPOSAL

– Must use AI/ML/LLM
– Should clearly benefit society or have commercial value



COMP3110 Accelerator | Project Proposal

In this form, you will submit a proposal for your Accelerator project in COMP3110.

Your accelerator project should be a useful, inventive, or delightful ML / LLM / AI -powered system. Social, commercial, creative, and everyday ideas are welcome. Consider who could benefit, including people whose needs are often overlooked.

Dr Jake Renzella, the LiC, will approve each proposal, or provide feedback. After approval, you may work solo or form a group and select one member's approved project.

Be concise and specific; bullet points are welcome. Suggested word limits guide scope. Unknowns are fine so long as you explain how you will investigate them.

Draft your answers before submitting. Deadlines and assessment details: https://comp3110.vercel.app/

Hi, Jake. When you submit this form, the owner will see your name and email address.

* Required

I. Submitter Info

Used to identify your individual proposal.

1. Student zID *

Enter your UNSW student ID, beginning with z.

Enter your answer

Next

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Microsoft 365

RESOURCES WE'RE MAKING AVAILABLE TO YOU

- CSE is providing access to AWS models on bedrock with a limited quote per student
- CSX.ai is providing $50 of highly optimised inference compute per-student
- LiteLLM can be used to manage API access to the variety of models
- You can also bring your own

RECAP: COURSE INTRO

1. MLE is the engineering around a model: problem, impact, deployment.
2. The term runs as an accelerator: teams, sprints, Demo Day.
3. 60% portfolio of tasks, 40% final project.

Next, 1.2: when is it time for AI.